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Artificial Intelligence

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BS (CS) Spring 2026

## Lab 07 Tasks



### Learning Objectives:

1. Local Search Algorithm
2. Simulated Annealing
3. Genetic Algorithm

# Lab Tasks

## Submission Instructions

1. Create one notebook file for all questions with name *ROLLNO\_SEC\_LAB7* e.g. **i23XXXX\_A\_LAB07**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

## Questions

**Q1.** Write a program to solve the **5-antenna** problem explained below using Genetic Algorithm. Randomly generate the population of 20 individuals/ chromosomes. Evaluate their fitness and then select, crossover and mutate. Run it for many generations/iterations until you find the individual with best/maximum fitness.

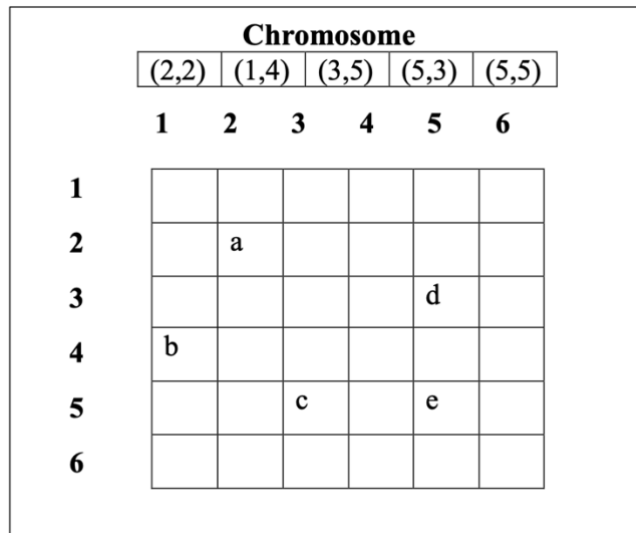
We must place 5 antennas (i.e. a, b, c, d, e) on a 6 x 6 grid in such a way that maximum area is covered by their signals. To find the best solution. We. Use Genetic Algorithm, upon a population of three individuals/chromosomes.

Follow the steps in the table below for the selection, mutation and fitness evaluation to find the best solution.

| Generation 1 | Selection and Crossover | Generation 2A | Mutation | Generation 2B | Selection and Crossover | Generation 3 |
|--------------|-------------------------|---------------|----------|---------------|-------------------------|--------------|
|--------------|-------------------------|---------------|----------|---------------|-------------------------|--------------|

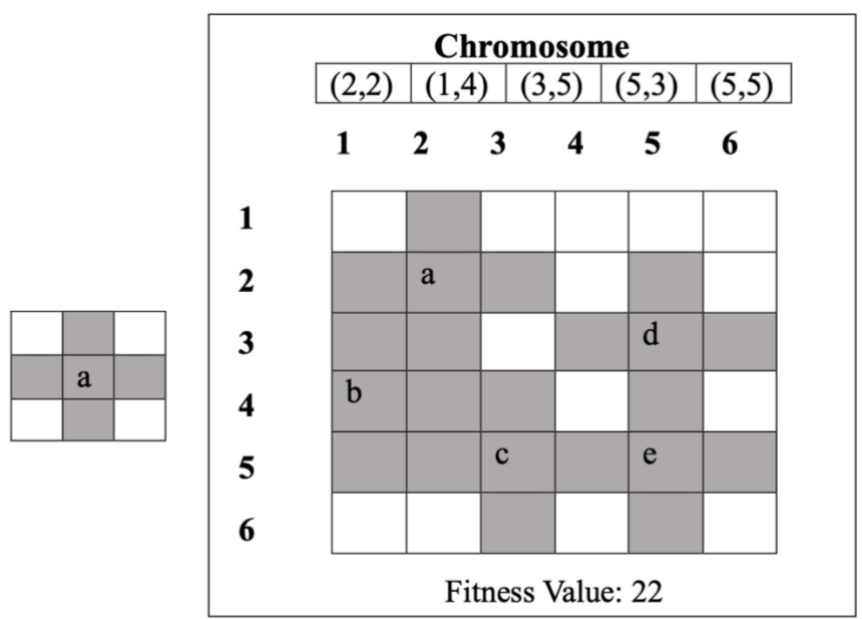
### Chromosome

The chromosome has 5 values, each representing the position of antenna (i.e. a, b, c, d, e) consecutively, on the grid. For example, the below chromosome shows the positions of a, b, c, d, e as shown. x is horizontal, and y is vertical.



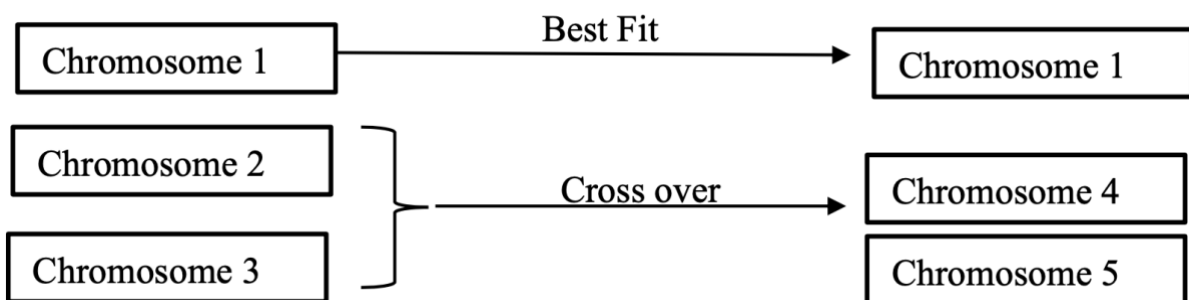
### Fitness Evaluation

Each antenna covers 5 blocks on a grid. One its own, and four neighbors, up, down, left and right. So, fitness value is the total area covered by all 5 antennas. Example is shown below. 5 antennas cover 22 blocks, so fitness value is 22.



### Selection

When the fitness value of all three chromosomes is calculated, we select the worst 2 chromosomes for cross over while the best chromosome passes to next generation as it is.

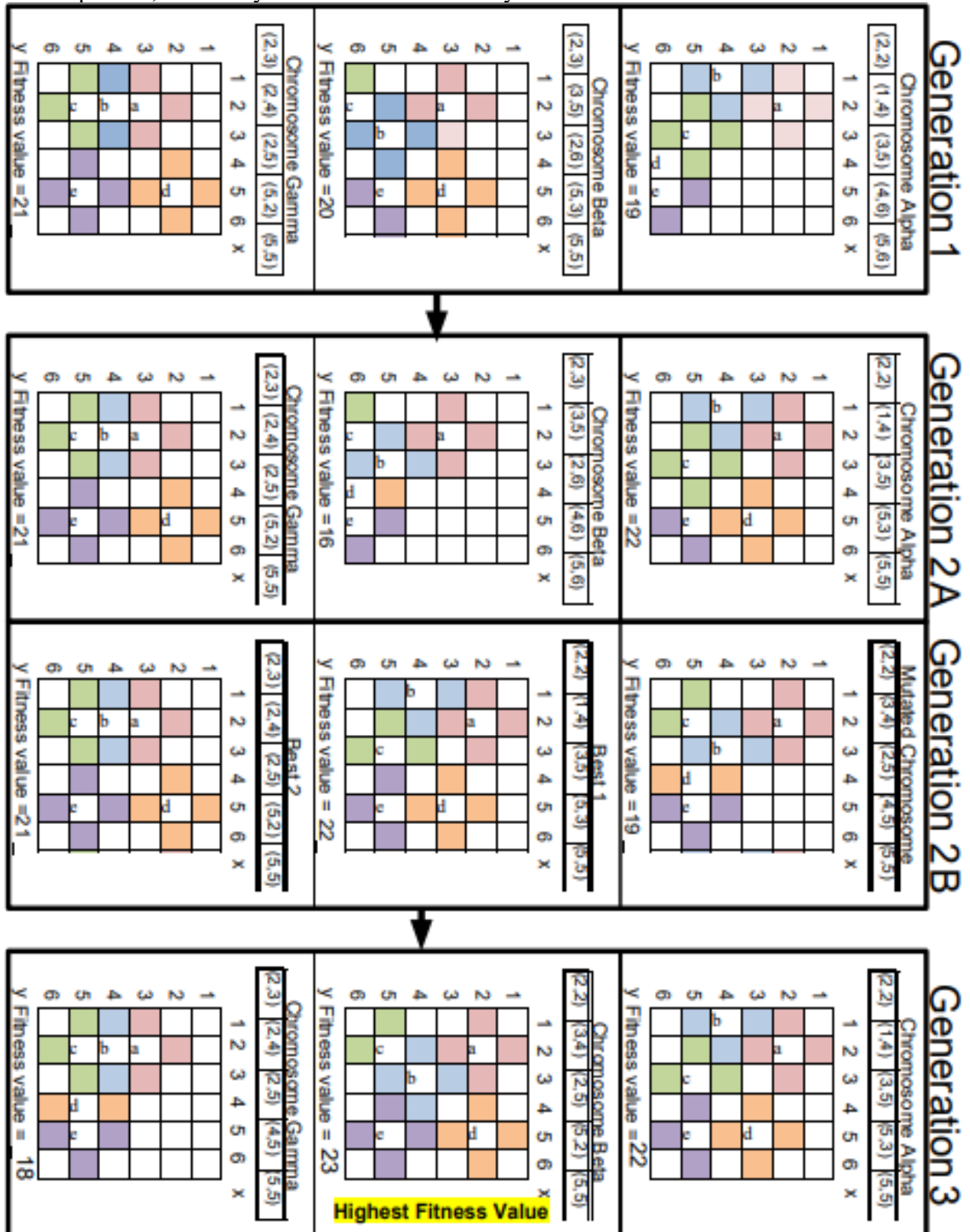


## Crossover

Let's example crossover takes place between sections (a,b,c) and (d,e), as shown in figure.

Mutation:

**ONLY** the worst chromosome is mutated out of the three. For mutation ALL the antennas, move upwards, i.e. their y value is decremented by 1.



Generic code structure: **Optional** to use this structure or made your own.

```

import random
def calc_fitness(chromo):
    # Calculate the fitness value for the chromosome
    # The fitness is the sum of the distances between each pair of
    # antennas # We want to maximize the fitness value

def selection(pop, fitness):
    # Use any selection criteria and justify why you use that
    # The chromosomes with higher fitness have a higher probability of
    # being selected

    return chromo1, chromo2
def crossover(chromo1, chromo2):
    # Perform crossover on the selected chromosomes to create a new
    # offspring
    return offspring1, offspring2

def mutation(chromo):
    # Mutate a chromosome return mut_chromo

# Main
function
def
main():
    # Create a random initial population of 20 chromosomes
    population = [[random.randint(0, 100) for i in range(5)] for j
in range(20)]
    print("Population:",population)
print('Highest Fitness value:',max(fitness),'at index',worst_idx1)
print('Fitness: ',fitness)
print('Chromosome 1: ',chromo1)
print('Chromosome 2: ',chromo2)

print('Offspring 1 After crossover and mutation: ',offspring1)
print('Offspring 2 After crossover and mutation: ',offspring2)
main()

```